Appendix D - Survey Instrument (Follow Up Survey)

The following is the survey content that was hosted on Qualtrics to facilitate user testing. Some elements of the survey (such as HTML elements, hyperlinks, and embedded videos, cannot be displayed in this document, and have been replaced with suitable alternatives.

Context

This is a survey collecting user feedback on an early prototype of a mobile quiz platform designed for use in primary schools. The platform is intended to support short, frequent revision sessions for students by using an easily accessible mobile app.

The survey will take approximately 20 minutes to complete and will require you to visit an external website as part of the evaluation. Please attempt this survey on a computer, as the website is not optimised for mobile or tablet devices. You may exit the survey at any time and return later. Your previous responses will be saved, but they will only be accessible if you continue the survey using the same device and browser. There are no right or wrong answers. Feedback on confusing or frustrating aspects of the platform is especially valuable.

Background

At which level(s) do you currently teach?

- ☐ Primary 1
- ☐ Primary 2
- ☐ Primary 3
- ☐ Primary 4
- ☐ Primary 5
- ☐ Primary 6
- ☐ Other: _____

How many years of teaching experience do you have?

- ◯ 0–2
- ◯ 3–5
- ◯ 6–10
- ◯ 10+

What subjects do you teach?

- ☐ Science
- ☐ Math
- ☐ English
- ☐ Other: _____

Scenario Based Usability

Introduction

The platform you are testing has two main interfaces: The teacher-facing web dashboard which facilitates the creation and assignment of quizzes to students, as well as the mobile application, which students will use to access and attempt the quizzes.

In the sections that follow, you will complete 3 scenarios, 2 of which require you to access the web dashboard. All tasks should be done on a computer as the site will not function properly on a mobile phone.

To better evaluate the usability of the platform, the scenarios will not include step-by-step instructions. We'd like you to explore the interface and figure out how to complete each task on your own. A tutorial is available for each scenario, but please only open it if you are genuinely stuck.

Scenario 1.1

Your teacher login credentials (Qualtrics webservice would send a post request to the credentials issuer here and replace the username and password fields below with the allocated credentials)

Username
${e://Field/teacher_user}

Password:
${e://Field/teacher_pass}

Scenario 1.1: Quiz Scheduling

Your task

Visit this (hyperlinked to webquizapp.delishad.com, where the prototype dashboard was hosted) webpage in a new window or tab, and sign in using the credentials provided above.

Look under the quizzes page (accessible from the side bar). There should be 2 existing quizzes

1. Water Cycle 1
2. Vocabulary 1

Look under the classes page (accessible from the side bar). There should be 2 existing classes

1. 6A
2. 6B

Picture yourself as a teacher who has created those two existing quizzes and would like to schedule the quizzes for both of your classes to attempt.

Schedule the quizzes for both classes with the following requirements.

- Each quiz should start today.
- Each quiz should end in 3 days.
- Each quiz should have 3 attempts allowed.
- Only proceed after the task has been completed.

View this tutorial only if you are unable to figure out how to do the task

There are actually many ways to schedule the quizzes. You can schedule them from the quiz page, the class page, or from the intended page.

The easiest way to schedule for multiple classes it through the scheduling page

Step 1: Navigate to the scheduling page
Step 2: Click on "+ Schedule New Quiz"
Step 3: Click the quiz you wish to schedule
Step 4: Select the classes and the dates for which the quiz will be assigned
Step 5: Adjust the attempts allowed to 3
Step 6: Assign the quiz.

The video below shows the scheduling of quizzes.

(Video is embedded in actual survey)

https://youtu.be/ahO6S63nkiU?si=rYDFpnT9XrhIHBDj

Scenario 1.2

Scenario 1.2: Quiz Scheduling

Your task

After assigning both quizzes based on the previous task, adjust the schedule assignment of the Water Cycle quiz for both classes.

- The quiz should now end 4 days from today (extended by 1 day)
- The quiz should now only have 2 attempts.

View this tutorial only if you are unable to figure out how to do the task

Step 1: Navigate to the scheduling page
Step 2: Right click the scheduled quiz that you would like to edit
Step 3: Adjust the dates and attempt count accordingly
Step 4: Save the changes

This video shows the adjusting of a class schedule.

(Video is embedded in actual survey)

https://youtu.be/RvAyuwmNUdw?si=kb-O6RWnyi8Xojv9

Scenario 1 Review

Please indicate your level of agreement with the following statements.

Statement	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
The location for scheduling a quiz was easy to identify.	☐	☐	☐	☐	☐
The scheduling process was easy to understand without external instructions.	☐	☐	☐	☐	☐
The interface guided me naturally through the scheduling process.	☐	☐	☐	☐	☐
Clear feedback was provided when quizzes were scheduled, moved, or updated.	☐	☐	☐	☐	☐
It was easy to adjust or correct a scheduling mistake.	☐	☐	☐	☐	☐
Rescheduling a quiz was straightforward.	☐	☐	☐	☐	☐
I felt confident that my changes to the schedule were saved correctly.	☐	☐	☐	☐	☐

What parts of the scheduling process felt unclear or unintuitive (if any)?

Are there any real classroom or school constraints that might make scheduling quizzes difficult using this system?

Scenario 2

Scenario 2: AI Quiz Generation

Your Task

The platform includes an in-built AI quiz generation tool. In this task, you will use the tool to generate quizzes for a topic of your choice.
Please choose a topic, subject, and education level that you are familiar with, so that you can better judge the quality of the generated quizzes.

Step 1: Default Settings

The following settings are already configured as the default values. You do not need to change them unless they have been modified.

Number of quizzes: 5
Questions per quiz: 10
Quiz Types: All Types
Timer Settings: Default

If you choose to generate Math quizzes, please remove the Crossword quiz type.

Step 2: Adjust the Other Options

You may choose/edit the following settings freely:

Education Level
Subject
Instructions
File Uploads (optional)

If you wish, you may upload up to 5 files to support the AI generation. File uploads are optional and are not required to complete this task.

Step 3: Generate and Review

Generate the quizzes using the AI tool.
Review the generated quizzes.

Important Notes

Please review your settings carefully before generating, as only one generation attempt is allowed.
Choose content that you are reasonably familiar with.
You do not need to edit or correct the generated quizzes.
Your goal is to use the feature and evaluate the output produced.

View this tutorial only if you are unable to figure out how to do the task

Step 1: Navigate to the quizzes page
Step 2: Click on "Generate Quizzes with AI"
Step 3: Select the relevant settings and type in your generation instructions
Step 4: Generate the quizzes and wait
Step 5: Review the generated quizzes

This video shows the generation of quizzes with AI

(Video is embedded in actual survey)

https://youtu.be/BNfw88cm3Ic?si=hDmkkBycAzekySiW

Scenario 2 Review

Please indicate your level of agreement with the following statements. (User Interface)

Statement	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
The location of the AI quiz generation feature was easy to identify.	☐	☐	☐	☐	☐
The AI generation interface was easy to understand without external instructions.	☐	☐	☐	☐	☐
The available generation settings (subject, level, instructions, etc.) were clearly presented.	☐	☐	☐	☐	☐
It was easy to configure the settings needed to generate a quiz.	☐	☐	☐	☐	☐
The process of generating quizzes using the AI tool was straightforward.	☐	☐	☐	☐	☐
The system clearly indicated when quiz generation was in progress or successful.	☐	☐	☐	☐	☐

Please indicate your level of agreement with the following statements. (Generated Output)

Statement	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
The generated quizzes followed the topic or instructions that I provided.	☐	☐	☐	☐	☐
The difficulty level of the generated questions was appropriate for the selected education level.	☐	☐	☐	☐	☐
The generated questions appeared factually correct.	☐	☐	☐	☐	☐
The answer options and correct answers were unambiguous.	☐	☐	☐	☐	☐
The questions did not feel repetitive.	☐	☐	☐	☐	☐

What parts of the AI quiz generation process felt unclear or unintuitive (if any)?

Were there any issues with the generated quizzes (e.g., incorrect answers, unclear questions, inappropriate difficulty)?

What improvements would you suggest for the AI quiz generation feature (if any)?

Scenario 3

Scenario 3: Student Facing Application (Gamification Elements)

You will now switch to the perspective of a student using the mobile platform.

Your task

Watch the video below that showcases the gamification elements of the student-facing mobile app. After you are done, click next and answer the subsequent questions.

Nothing else needs to be done for this task besides watching the video.

(Video is embedded in actual survey)

https://youtu.be/5ymaSw4bU0Q?si=x8KtsGFNOCEenTsp

Scenario 3 Review

Please indicate your level of agreement with the following statements.

Statement	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
The avatar customisation system would motivate students to participate more consistently.	☐	☐	☐	☐	☐
The leaderboard system would encourage students to participate more consistently.	☐	☐	☐	☐	☐
The badge system would motivate students to participate more consistently	☐	☐	☐	☐	☐
Overall, the reward systems shown would make quiz practice more engaging for students.	☐	☐	☐	☐	☐
The gamification elements shown are appropriate for primary school students.	☐	☐	☐	☐	☐
The competitive elements (leaderboards) are suitable for a classroom environment.	☐	☐	☐	☐	☐

Do you think the gamification features shown would motivate students to complete quizzes regularly? Why or why not?

Do you have any concerns about the gamification features shown?

What improvements or additional features would you suggest for the gamification system (if any)?

Concluding Remarks

Any other feedback